

Can microscope photos tell these battery batches apart?

34 BSE micrographs · 3 curated batches + 3 new photos · frozen recipe fitted on Batch 3 · generated 2026-10-04

We measured every photo the same way and asked three questions: do the batches hold the same amounts, do they differ in structure, and can we prove it's the material — not the imaging — that differs? The amounts match; the imaging sessions don't. Today the honest answer for every batch is CAN'T TELL YET — not because nothing differs, but because session and batch can't be separated in this dataset.

BATCH	VERDICT	CONFIDENCE	MAIN REASON	NEXT STEP
Batch 1	CAN'T TELL	low	differences track imaging session, not batch	interleaved re-imaging
Batch 2	CAN'T TELL	low	no difference above detection limit; sessions still entangled	same re-imaging
New photos (3)	CAN'T TELL	low	0 of 2 confirmed; calls matched sessions	blind re-test with session control

How to read this report

Each dot is one microscope photo. The grey band is the normal range seen in Batch 3, the approved reference batch — dots inside it look normal. Every claim carries a label:

SOLID the measurement is consistent and was checked **POSSIBLE** the pattern is real but confounded **CAN'T TELL YET** the data cannot decide

What we measured, and how



The same ruler measures all 34 photos — it was set once on the approved Batch 3 and never re-fitted, so batches can't drift the ruler. Below, one photo before and after colouring:

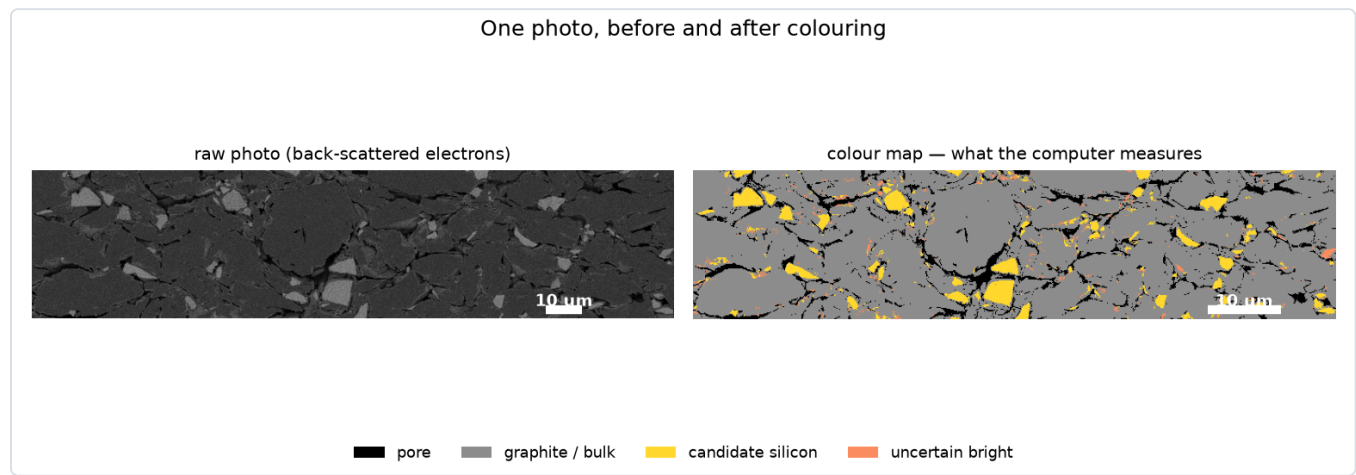


Figure 1. Left: a typical photo. Right: the same photo coloured into pore (black), graphite bulk (grey), candidate silicon (yellow) and uncertain bright material (orange). Bar = 10 μm .

What we found

SOLID

1. All three batches contain the same amounts of pore space and silicon.

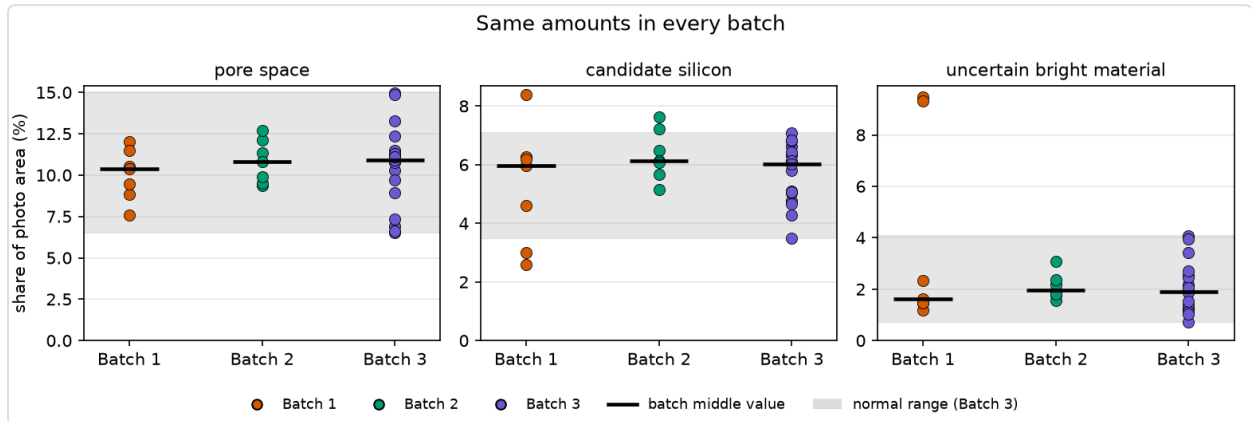


Figure 2. Same amounts in every batch. Each dot is one photo; the short line is the batch's middle value; the grey band is the normal range seen in Batch 3, the approved batch.

WHAT WE FOUND

Pore space is about 10%-11% of each photo and candidate silicon about 6%, in all three batches. Every batch's middle value sits inside Batch 3's normal range.

HOW SURE WE ARE

Solid for differences bigger than about 1.4 percentage points of silicon or 2 points of pore space. Those are the smallest gaps 7, 7 and 17 photos can reveal (exact test).

WHAT IT DOES NOT MEAN

It doesn't prove the batches are identical. Smaller differences, or material too fine to resolve at 25 nm per pixel, would not show up.

Source: validated_comparison/per_image_features.csv, pairwise_comparisons.csv

POSSIBLE

2. Two Batch 1 photos contain unusual bright material whose identity we can't tell.

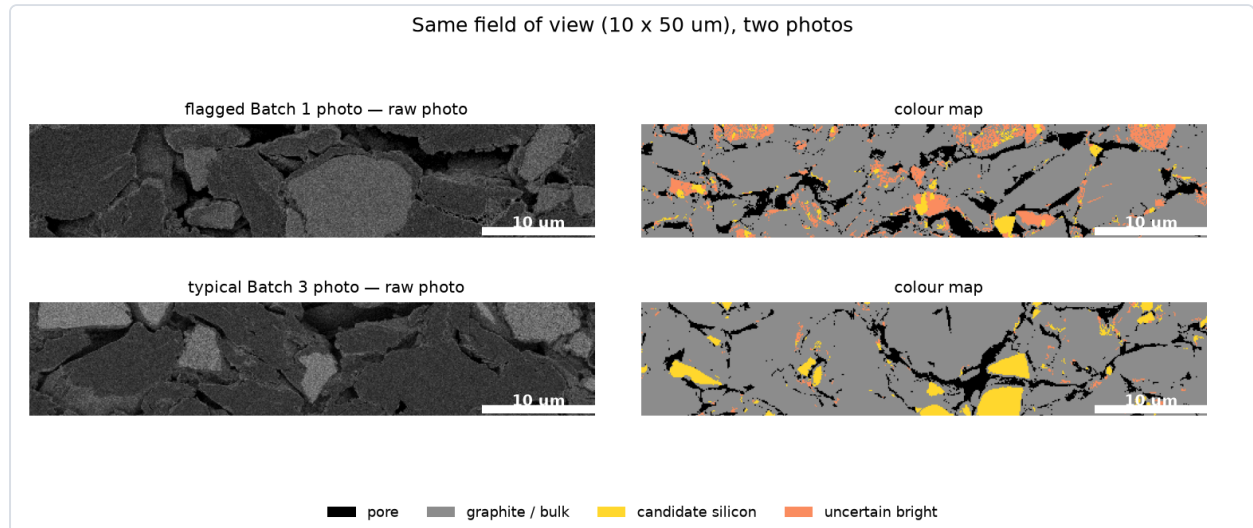


Figure 3. Unusual bright material, up close. The flagged Batch 1 photo (top) carries far more orange 'uncertain bright' material than a typical Batch 3 photo (bottom), at the same 10 x 50 μm scale.

WHAT WE FOUND

In those two photos, uncertain-bright material covers 9% and 9.3% of the area — more than double the most seen anywhere in Batch 3 (4.1%). It is genuinely bright material, not grey bulk we misread.

HOW SURE WE ARE

The amount is measured and solid. What the material is made of is not established: brightness is not chemistry, so this is labelled uncertain rather than called silicon.

WHAT IT DOES NOT MEAN

It doesn't prove a defect. It is a photo-level anomaly that could be imaging, preparation or real material — the next step is an EDS elemental scan on that material.

Source: image_acquisition.csv, objects_classified.csv

SOLID (as an observation)

3. The photos were taken under different imaging conditions that line up with the batches.

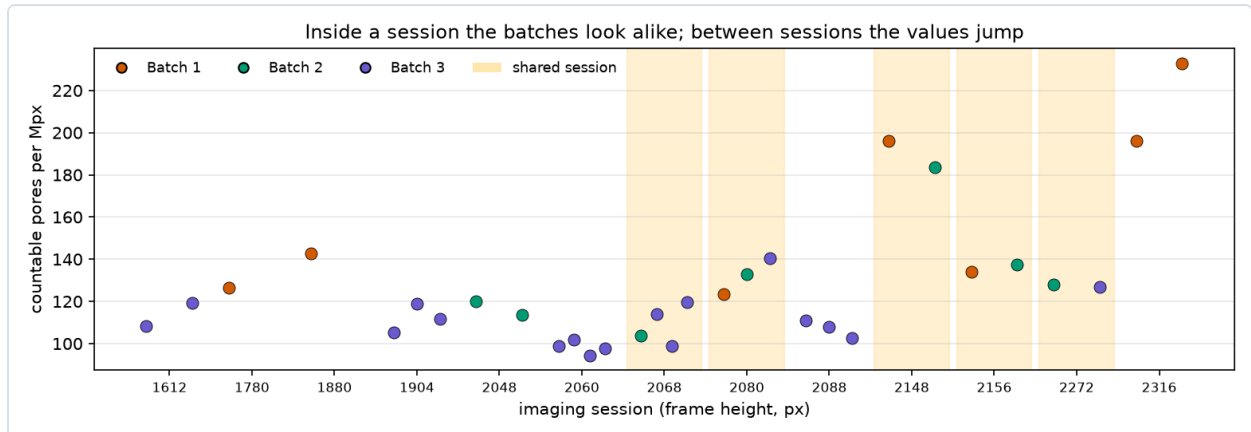


Figure 4. Sessions, not just batches. Each dot is one photo's countable pores; columns are inferred imaging sessions (frame height); dot colour is the batch; amber columns hold more than one batch.

WHAT WE FOUND

Inside a shared session, photos from different batches look alike; between sessions, values jump. A classifier using only acquisition signature (frame shape, noise, contrast) matches batch labels 61% of the time — as well as the 59% achieved using the measured structure.

HOW SURE WE ARE

Solid as an observation: the session pattern is in the images themselves (13 inferred sessions; 21/34 photos have the darkest pixels clipped to pure black).

WHAT IT DOES NOT MEAN

It doesn't prove the batch labels are wrong — it proves we can't yet separate material from how the photos were taken.

Source: image_acquisition.csv, benchmark_results.csv

CAN'T TELL YET

4. So batch differences can't be confirmed yet.

BATCH PAIR	SHARED SESSIONS	VERDICT
Batch 1 vs Batch 3	1	untestable — one shared session
Batch 1 vs Batch 2	3	inconclusive
Batch 2 vs Batch 3	3	inconclusive

WHAT WE FOUND

The decisive test is comparing batches inside the same session, where imaging conditions match. Batch 1 and Batch 3 share only one session — that is too few photos to call a difference.

HOW SURE WE ARE

With 13 sessions spread unevenly over 31 photos, the controlled comparison lacks the numbers it needs. This is an absence of evidence, not evidence of absence.

WHAT IT DOES NOT MEAN

It doesn't say the batches are the same. It says this dataset cannot currently separate material differences from session effects.

Source: blocked_permutation.csv, image_acquisition.csv

CAN'T TELL YET

5. The new photos: what our blind calls taught us.

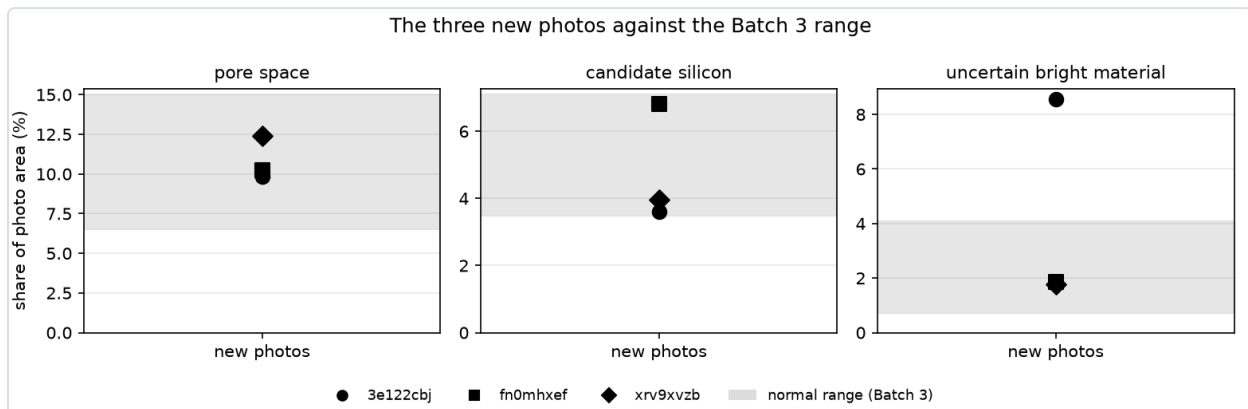


Figure 5. New photos against the normal range. The three new photos' area shares against the Batch 3 normal range. One photo (3e122cbj) sits far outside on uncertain bright material.

PHOTO	OUR SAVED BLIND CALL	CONFIRMED ANSWER	NOTE
img_3e122cbj	Batch 1	Batch 2	wrong — call followed its session
img_fn0mhxef	Batch 2	Batch 1	wrong — call followed its session
img_xrv9xvzb	Batch 3	Batch 3	consistent, unconfirmed

WHAT WE FOUND

We matched each photo to the closest batch before any answers were known, and saved the calls. Confirmed corrections make it 0 correct out of 2 (the third is unconfirmed). Both wrong calls matched the photo's imaging session — the exact confound in finding 3, flagged on the calls themselves.

HOW SURE WE ARE

The labels are externally confirmed by the organizer; our calls were timestamped before disclosure. The score is honestly 0/2.

WHAT IT DOES NOT MEAN

It doesn't prove the pipeline sees nothing — it proves that on these three photos, session signature and batch pointed the same way, so a correct call wouldn't have meant material was recognised either.

Source: new_batch_ground_truth.csv, assignments.csv

What this means for a QC decision

No measured difference survives the session control, so the evidence does not support REJECT for any batch. It also doesn't support 'identical material': the dataset cannot currently separate the two. The useful actions are about removing the confound:

- Get real session metadata (or re-image the batches interleaved in one session) — the single fix that unlocks the batch question.
- EDS elemental scan on the uncertain bright material in the two flagged photos.
- One measured bulk porosity per batch — anchors the image-based pore share.
- Repeat imaging of the same material — separates photo noise from manufacturing change.

Limitations

- Photos are crops from ~15 parent images, not independent specimens — batches are curated, not production lots.
- No crop-to-parent map exists, so independence and generalization can't be tested.
- 'Silicon' is a brightness label — only EDS confirms chemistry.
- 2-D sections can't show 3-D connectivity; small 2-D profiles aren't small particles.
- Below-resolution detail (<25 nm/pixel) is invisible by construction.
- The three new photos are unblinded — future method changes need a fresh test set.

Appendix

Measurements we left out, and why

MEASUREMENT	WHY IT IS NOT USED FOR DECISIONS
pore counts, sizes and shapes	pore count is the only value that passes the many-test correction, but the session check can't run (one shared group); sizes carry the section-cut caveat
clustering and connectivity	sensitive to where boundaries land; nothing survives the session check
silicon–pore contact	corrected and real, but not separable from session effects
battery-model (DFN) indicators	model-derived; the differences sit inside the model's own assumption spread

extra detector channels (InLens/ETD)	exploratory; detector gain varies with session
orientation strength (S)	real shared structure — pores lie near-horizontal in all batches — but it describes the electrode rather than separating batches

Glossary

resolved pore	dark regions large enough to measure (>0.13 um across)
candidate silicon	bright, compact objects that look like silicon — a label, not a chemical test
uncertain bright	bright material that doesn't look like solid grains — thin or edge-rich; kept as its own label instead of guessing
session	a group of photos that look like they were taken together (same frame size, detector set, noise)
smallest detectable difference	the smallest shift this many photos could reveal — anything smaller is invisible to the test
matched call	assigning a photo to the batch it most resembles

Data and reproducibility

One command rebuilds everything from the saved tables: `python build_report.py`. Frozen-recipe hash: `039bob88a611...` (`dfn_output_v2/run_manifest.json`). The '34 micrographs' count is the files on disk; 'curated batches' is the organizer's description of the dataset.